

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

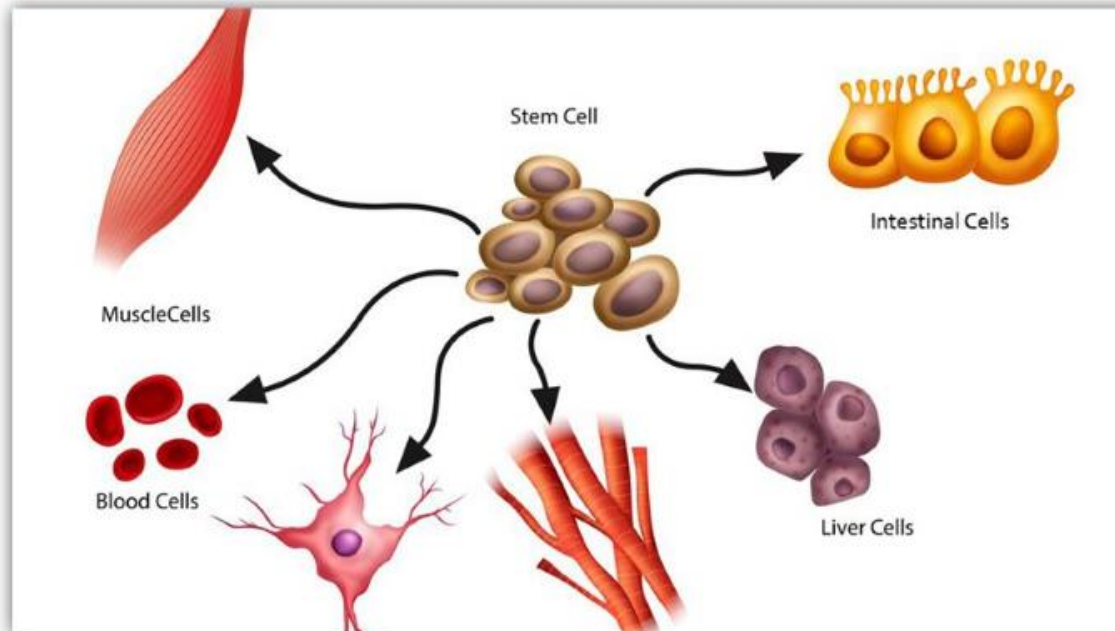
MODULE - 18

BIOTECHNOLOGY - 8

BIOTECHNOLOGY

STEM CELLS

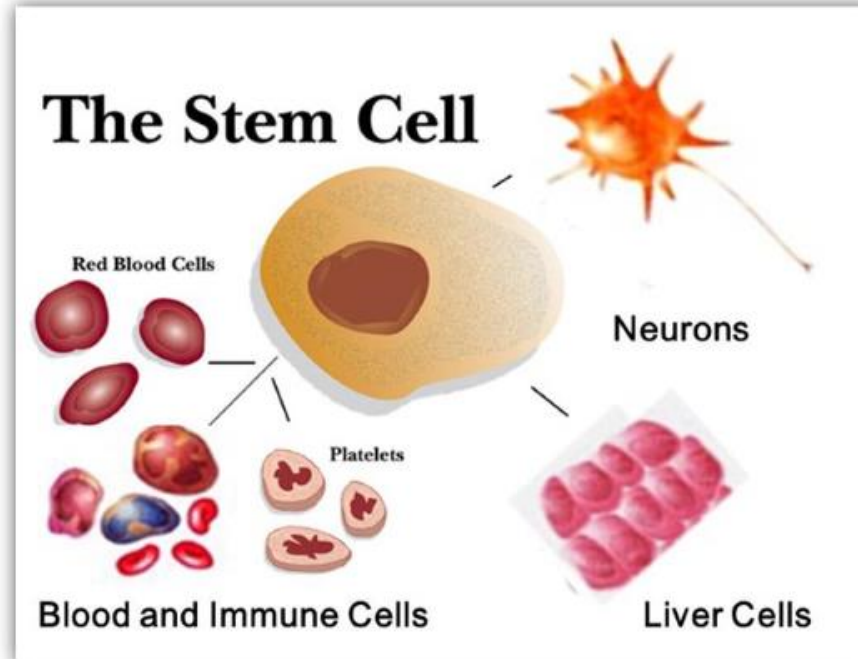
- We humans reproduce sexually, the sperm fuses with egg to form a fused Zygote. The single celled Zygote divides itself to develop Foetus.
- So, a single celled Zygote develops the entire human body having all the organs.
- So, one cell is capable of producing an organism. This is called Totipotency, which means the ability to develop a complete organism.



STEM CELLS

WHAT ARE STEM CELLS?

- Stem cells should be
 - i. Able to divide and differentiate themselves into specialized cells.
 - ii. Should have the capability of cell division (mitosis) to produce more cells.



WHERE ARE STEM CELLS FOUND?

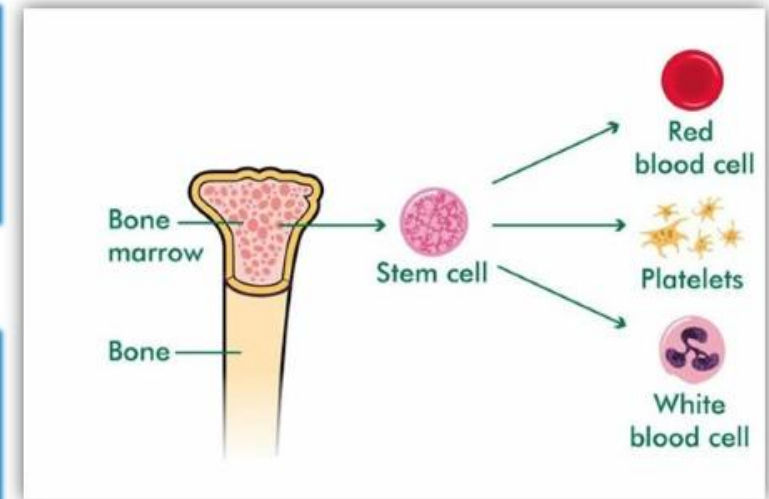
- The human embryo is a good source where the cells are totipotent.
- Stem cells are also present in adults (as well as in children).
- Adult stem cells or Somatic cells or Non-Embryonic cells are found in bone marrow and other organs (heart, brain, kidney and gut). They are found as unspecialized cells among the specialized cells in tissues and organs.
- These cells carry out the important function of replenishing tissues (replacing the dead cells).
- They have the ability to divide infinitely to generate all the cell types of the organ from which they originate. So, they can potentially regenerate the entire organ from a few cells.
- So, they serve as a sort of internal repair system, dividing without limit to replenish cells in the tissues.
- In some organs like the gut and bone marrow, stem cells regularly divide to repair and replace worn out and damaged tissues.

STEM CELLS**PROPERTIES OF STEM CELLS**

1. Stem cells are capable of dividing and renewing themselves.
2. They are unspecified/undifferentiated.
3. They can give rise to specialized cell types.

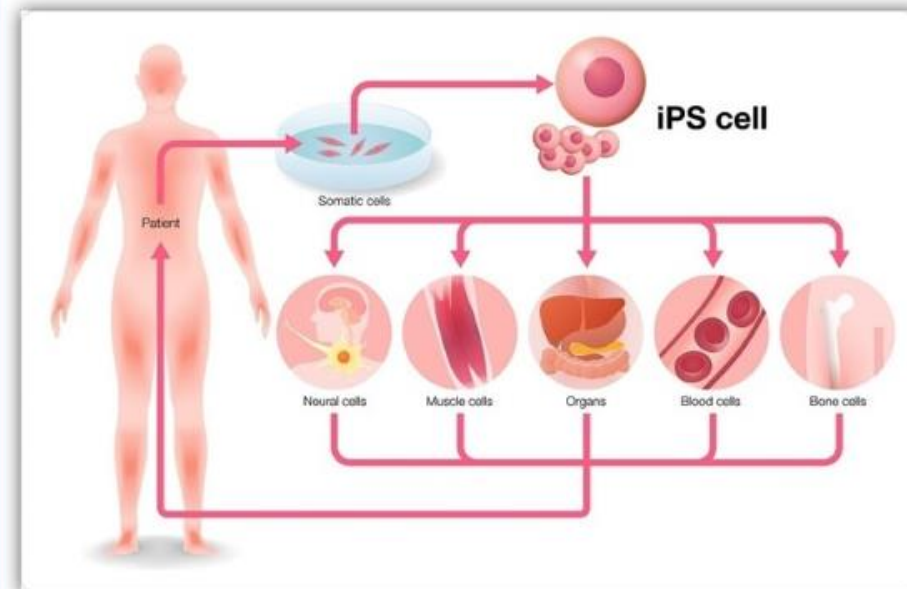
APPLICATIONS OF STEM CELLS

1. The main application of Stem cells is to replace dead cells and tissues destroyed by diseases like blood cancer, Parkinson's disease etc.
2. Currently, the blood forming Adult Stem cells of bone marrow are the only type of stem cells in use for treating diseases.
3. Stem cell therapy can be used as a cure for Type -1 Diabetes.
4. Stem cells can be used in organ transplants, they can be used for burn victims for skin replacement, to treat spinal cord injuries and to treat eye diseases.
5. Teeth and hair can also be grown from stem cells.



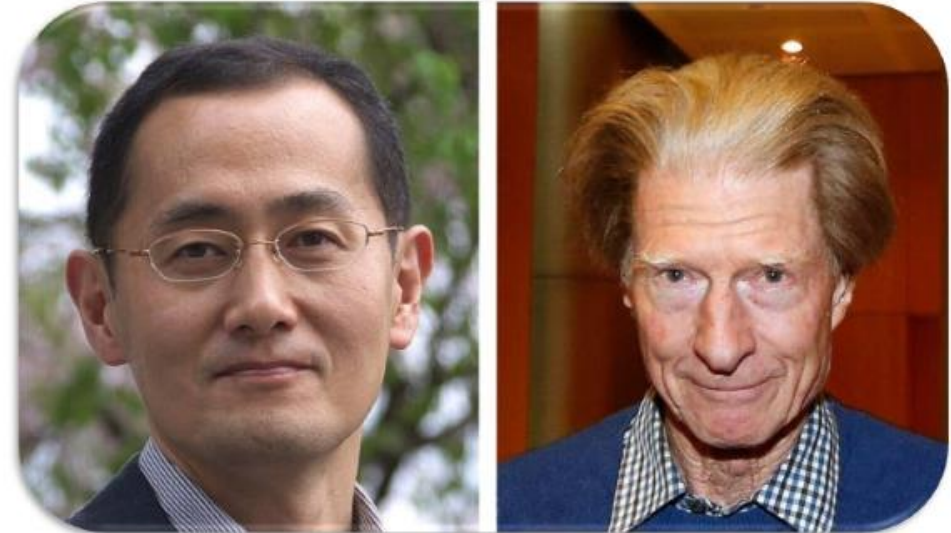
CONTROVERSIES SURROUNDING STEM CELL THERAPY

1. Embryonic stem cells must be obtained when an embryo is in early stage of development. The embryonic cells are separated and maintained in a lab thereby halting embryonic development.
2. Many people view this as killing of child. Legally embryo the day it is formed is considered as human life and it enjoys all the rights like Article 21.
3. Also, this can lead to development of a black-market for embryos and poor people would be exploited.
4. Using Adult Stem cells and Induced Pluripotent Stem Cells (IPSC) poses less ethical dilemma. But extracting Adult Stem cells is very difficult.



INDUCED PLURIPOTENT STEM CELLS (IPSC)

1. Undifferentiated stem cells turn into differentiated cells. But by genetic modification of the differentiated cells, they can be turned into IPSC.
2. IPSCs are like the Embryonic Stem cells.
3. IPSCs share many characteristics with the human Embryonic Stem cells (Totipotency).
4. There are many more differentiated cells in our body than undifferentiated cells. So, a large number of undifferentiated cells can be made from a patient's own differentiated cells (like skin cells) instead of destroying an embryo to produce Embryonic Stem cells.
5. For this discovery John B Gurdon and Shina Yamanaka got the Nobel prize in medicine in 2012.



B GURDON AND SHINA YAMANAKA

OTHER ISSUES

1. The iPSC technology is not still fully functional. Obtaining full functionality is still a major challenge.
2. Stem cell therapy is not an inclusive treatment. It is very expensive, so poor people are not covered,
3. There is a scope of misuse.



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